

HUMAN FLAG ASSOCIATION

Bellinzona, Switzerland · UNGM Partner No. 6126 · humanflag.org

Technical Note HFA-TN-02

The Reciprocal Time Window as a Precondition of the Recognition Obligation

Article 41 of Additional Protocol I, customary international humanitarian law, and the temporal preconditions of recognising expressed intention

5 August 2026 · CC BY 4.0 · DOI: [10.5281/zenodo.21224900](https://doi.org/10.5281/zenodo.21224900)

Giovanni Nardacci, Human Flag Association

Thesis. Recognition of intention is a function of the reciprocal time window afforded to the exchange. Because the obligation to recognise a person *hors de combat* already exists — under customary international humanitarian law, binding on all parties to armed conflict, and as codified in Article 41 of Additional Protocol I — the minimum window that makes recognition possible is the structural precondition of that obligation’s exercisability — of the practical ability to comply with it in the circumstances of use. Whether a system preserves that precondition is capable of verification, and must be examined in the applicable legal review of the weapon — including under Article 36 of Additional Protocol I where that provision applies, and under the corresponding national review requirements where it does not. Compressing the window below the threshold of exchange by design is therefore an examinable design decision, not a technical circumstance.

1. Purpose and placement

This note completes a documented progression across three prior instruments. First, the publication of HF SIGNAL 01 — a dual-channel surrender-recognition specification (7 Hz pulsating white light; 144.0 MHz FM tone) — removed the categorical claim that no machine-readable surrender signal can be publicly defined: such a signal exists and is publicly specified. It does not, and does not purport to, establish operational feasibility in every environment; occlusion, interference, spoofing, sensor performance and classification remain open engineering questions. Second, the accompanying working paper (“Paper III”) established the declared-choice thesis: once the specification is public, non-recognition ceases to be a limitation and becomes an examinable decision. This note takes the third step: where the legal obligation to recognise already exists, the examination has a defined object. The time window that makes recognition possible is the structural precondition of the obligation’s exercisability, and the preservation of that precondition is a matter for legal review.

This note does not repeat the argument of Paper III. Paper III addressed the status of non-recognition (limitation vs. decision). This note addresses the status of the time window itself (technical circumstance vs. reviewable design parameter). The two are cumulative, not duplicative.

2. Recognition as a reciprocal act

The element introduced here, and absent from prior instruments, is reciprocity. Recognition of intention is not a unilateral reading of a signal. It is a minimal exchange with at least three temporally ordered components: (i) emission of the signal by the party expressing the protected intention; (ii) perception and evaluation of the signal by the opposing system or operator; (iii) the translation of that evaluation into attack inhibition — the withholding or aborting of the engagement — before the point of irreversibility. Each component consumes time. The exchange therefore has an irreducible minimum duration: an end-to-end interval running from emission to inhibition. “Reciprocity” here

does not denote reciprocity of legal obligation — surrender is a unilateral manifestation to which a duty of the opposing party corresponds. It denotes the two-sided temporal structure of expression and recognition: the interval must accommodate two chains, not one — the chain by which the person produces a perceptible manifestation, and the chain by which the opposing system perceives it and acts upon it. The point of irreversibility is understood throughout as a property of the causal sequence itself — the point beyond which the outcome can no longer be altered by any intervention — and not as a function of any capacity to respond to it (see “Systemic Arrestability”, Section 3.2).

A fourth element — a response perceivable by the person expressing the intention, allowing that person to know the signal has been received — completes the exchange in operational terms and is decisive for the reliability of the interaction. It is not asserted here as a precondition of the legal obligation, which is discharged by the withholding of attack itself; it is retained as a design and reliability requirement. The legal argument of this note rests on the end-to-end segment alone.

From this follows the central distinction of the note: below the minimum duration of exchange, recognition is not missed — it is precluded. A missed recognition is a failure within a process that could have succeeded. A precluded recognition is the elimination of the process itself. The legal consequences differ accordingly.

The relevant threshold is not a universal number of milliseconds. It is a scenario-specific end-to-end interval sufficient for a protected intention to be expressed, detected, evaluated and translated into attack inhibition, with the required degree of reliability, before the engagement becomes irreversible. Its adequacy depends, at least, on the duration and observability of the signal; sensor coverage and performance; detection and classification latency; the confidence threshold applied; the time needed to inhibit or abort the effector; and the effector’s point of irreversibility. The threshold is therefore not a tautology but a measurable, verifiable parameter.

Measurability, however, carries a condition. Where a system’s behaviour is not stable over its operational life — as with systems that adapt, learn, or are updated after review — the interval is not a single value but a distribution which may shift after the measurement has been taken. Two consequences follow. The parameter to be recorded is the reasonably foreseeable adverse case within the system’s validated operational envelope, rather than the typical case; and the determination cannot be a single act, since a system whose behaviour has changed is a system no longer verified in the respect that matters. The requirement, in the machine safety context, that systems exhibiting self-evolving behaviour remain correctable at all times in order to maintain intrinsic safety points in the same direction.

3. The window as precondition of exercisability

Article 41 of Additional Protocol I prohibits attacking a person who is recognised or who, in the circumstances, should be recognised as *hors de combat*. The ICRC Commentary on the Additional Protocols (1987) reads the second limb through the standard of the reasonable combatant acting in good faith in the circumstances ruling at the time: what “should have been recognised” is assessed against what such a person could perceive. That standard presupposes that perception remains possible. An engagement cycle compressed below the minimum duration of exchange does not merely raise the evidentiary bar of the standard; it removes the factual substrate on which the standard operates.

This note therefore does not claim that the time window is itself part of the treaty content of Article 41. It claims something narrower and structurally stronger: the window is the precondition without which the obligation cannot be exercised in practice — and

preconditions of exercisability are precisely what a legal review of the weapon can examine. A party that designs, procures, or deploys a system whose engagement cycle forecloses the exchange does not escape the norm; it renders its own compliance unverifiable by design.

Two objections are anticipated here, because they mark the limits of the claim. The first is that Article 41 prohibits attacking a person who has surrendered but imposes no duty to slow an attack so that an adversary may surrender; on that reading, no obligation to afford a window can be derived from it. The objection is well taken against a claim this note does not make. Nothing here asserts a right to a pause. The distinction is between a surrender manifested too late in a given engagement — a failure within a process that could have succeeded — and a system configured so that no otherwise timely manifestation, made within the intended circumstances of use, can be recognised before irreversibility. The first is a contingency of combat. The second is a property of the design, and it is the second alone that this note places before legal review.

The second objection is the mirror of the first, and is the more serious: that where acceptance of surrender is not feasible in the circumstances, the surrender does not bind the attacking party, so that a compressed cycle defeats the surrender rather than exposing a defect in the system. This is correct as to a particular engagement and cannot be extended beyond it. Feasibility is assessed in the circumstances ruling at the time; it is not a licence to constitute those circumstances in advance. A party that fixes, at the design stage, the conditions under which acceptance will never be feasible has not encountered an infeasibility — it has produced one. The question for review is accordingly not whether acceptance was feasible on a given occasion, but whether the system as designed leaves any occasion on which it could be.

There is, moreover, a second route by which the temporal interval enters the legal review, and it does not derive a right to a pause from Article 41. Article 57(2)(b) of Additional Protocol I requires that an attack be cancelled or suspended if it becomes apparent that the objective is not a military objective, is subject to special protection, or that the attack would be disproportionate; a corresponding obligation, there qualified by feasibility, is recorded as customary in Rule 19 of the ICRC study, on practice extending to States not party to the Protocol. That duty presupposes something about the engagement it governs: that where legally relevant information becomes available before the point of irreversibility, the architecture has not already made cancellation or suspension impossible by design.

The proposition this establishes is narrow, and it is worth stating what it is not. It does not require that every weapon remain technically recallable after release, nor that any engagement be interruptible on demand. It is this: where the intended circumstances of use permit a legally relevant change of status to become apparent while the engagement is still causally alterable, the review must examine whether the system preserves any effective pathway by which that information can result in cancellation, suspension, or referral for further human judgment. The duty to cancel or suspend is not specific to surrender: it applies whenever the legally relevant conditions identified by the rule become apparent. The interval it presupposes is therefore not created for the benefit of a surrendering adversary; the recognition of surrender is one of the cases it serves.

National instruments give this requirement specific operational form. DoD Directive 3000.09 requires that autonomous and semi-autonomous weapon systems be designed to allow commanders and operators to exercise appropriate levels of human judgment over the use of force, and that they complete engagements within a timeframe and geographic area, and other relevant environmental and operational constraints, consistent with commander and operator intentions; where a system is unable to do so, it is to terminate the engagement or obtain additional operator input before continuing (§

1.2.a.(1)(b), reprised among the matters to be verified before formal development at § 4.1.c.(2)). The same directive requires clear procedures for trained operators to activate and deactivate system functions (§ 1.2.a.(3)(c)), and states as an ethical principle of governability the ability to disengage or deactivate deployed systems that demonstrate unintended behaviour (§ 1.2.f.(5)). These provisions do not establish a general entitlement to interrupt any engagement on demand. They establish that the continuation of an engagement is conditioned in advance on temporal and operational constraints, and that a system unable to remain within them is to stop or defer to a human.

Where such a pathway is relied upon, it must be genuinely exercisable. A nominal halting pathway that another co-deployed system can tamper with, falsify or circumvent does not preserve the temporal opportunity it purports to provide; arrestability accordingly falls to be assessed at the level of the system of systems rather than agent by agent (see HFA Technical Note TN-01, 'Third-Party Interference with Shutdown Mechanisms', 5 July 2026).

The scope of this route should not be overstated. A halting capacity supplies time; it does not supply perception. That a system can be stopped within the cycle does not entail that a manifestation occurring within that cycle will be detected, classified and acted upon: those remain distinct requirements, and this note does not derive them from arrestability. What the route establishes is narrower and sufficient for the argument made here — that the temporal interval whose preservation is at issue is not an indulgence sought on behalf of an adversary, but a property the system is required to have for reasons of control that the deploying party has already accepted. Where the interval is owed in any event, the question of what can be recognised within it is properly before the review.

The consequence for Article 36 is direct but conditional in form. Article 36 of Additional Protocol I requires a State, in the study, development, acquisition or adoption of a new weapon, means or method of warfare, to determine whether its employment would, in some or all circumstances, be prohibited by applicable international law. Article 36 does not prescribe a particular temporal test; it prescribes the determination. Where, however, the intended circumstances of use permit a person to become *hors de combat* during the engagement cycle, that determination cannot be completed without assessing whether the system preserves a sufficient opportunity for the protected status to be expressed, recognised and translated into attack inhibition before the engagement becomes irreversible — what end-to-end interval the system affords, under what conditions, and how it is measured and recorded. A review that does not examine this question has not examined whether compliance with the recognition obligation remains possible in the circumstances of intended use. What the examination requires is that the question be reached, not that it be answered in a particular way: its outcome may be a design safeguard, an operational restriction, or the exclusion of particular circumstances of use from the authorisation.

The argument does not depend on ratification of Additional Protocol I, and it can be stated entirely from within the legal architecture of a State not party to it. Such a State — the United States is the principal example — is bound by Article 23(c) of the Regulations annexed to the 1907 Hague Convention IV, which it ratified in 1909 and which prohibits killing or wounding an enemy who, having laid down his arms or having no longer means of defence, has surrendered at discretion. Its own military manual recognises the protection of persons *hors de combat* — including those who surrender — as binding law (U.S. Department of Defense, Law of War Manual, § 5.9; on surrender, § 5.9.3), and the customary character of the rule is recorded in the ICRC Customary IHL Study, Rule 47.

Such a State likewise conducts legal reviews of new weapons under directives of its own issuance: DoD Directive 5000.01 requires legal review of the acquisition of weapons and weapon systems, and DoD Directive 3000.09 requires that autonomous weapon systems

be reviewed against national and international law, including the law of war. For that State, the verification identified in this note is therefore not an external requirement derived from a treaty it has declined to ratify: it is the operational consequence of obligations it has itself written and never disavowed. The question is identical in either architecture: does the engagement cycle preserve the exchange that the State's own rules presuppose?

This note is the perceptual-relational counterpart of a parallel criterion already deposited: systemic arrestability (Zenodo, DOI 10.5281/zenodo.20837150), which requires that a system's capacity to halt a hazardous action not depend on human perception, evaluation, and intervention within the action's time window. Arrestability governs the capacity to stop; the reciprocal window governs the capacity to recognise. Both are temporal criteria attached to obligations already written; neither creates new law.

4. Operative conclusion

The question this note places before any State operating or reviewing autonomous or automated engagement systems is the following: if the obligation of recognition presupposes a minimum reciprocal time window, and that window is neither guaranteed by design nor verified in the legal review of the weapon, on what basis is compliance asserted? The note does not assert that it cannot be. It records, on a dated instrument, that the question has been asked. The date of deposit records the public articulation of a specific, testable compliance question; it does not create the underlying obligation. The deposit does not alter the law and does not put any State on constructive notice. It provides a dated, public and citable articulation of the proposition that compression of the window is an examinable design parameter rather than a merely technical circumstance.

An empirical demonstrator of the temporal dependence of recognition — showing that varying the window alone, all other variables held constant, alters the outcome of the recognition process — will be the object of subsequent work.

References

Protocol Additional to the Geneva Conventions of 12 August 1949, and relating to the Protection of Victims of International Armed Conflicts (Protocol I), 8 June 1977 — Articles 36, 41 and 57.

Regulations concerning the Laws and Customs of War on Land, annexed to Hague Convention (IV), 18 October 1907 — Article 23(c).

Sandoz, Y., Swinarski, C., Zimmermann, B. (eds.), Commentary on the Additional Protocols of 8 June 1977 to the Geneva Conventions of 12 August 1949, ICRC/Martinus Nijhoff, 1987 — commentary on Articles 36 and 41.

Henckaerts, J.-M., Doswald-Beck, L., Customary International Humanitarian Law, Vol. I: Rules, ICRC/Cambridge University Press, 2005 — Rules 19 and 47.

U.S. Department of Defense, Law of War Manual (June 2015, updated July 2023) — § 5.9 (Persons Placed Hors de Combat); § 5.9.3 (Persons Who Have Surrendered).

U.S. Department of Defense, Directive 5000.01, The Defense Acquisition System — legal review of the acquisition of weapons and weapon systems.

U.S. Department of Defense, Directive 3000.09, Autonomy in Weapon Systems.

Human Flag Association, "Systemic Arrestability", Zenodo, DOI 10.5281/zenodo.20837150.

Human Flag Association, "Lawful Operational Safeguards in AI Systems", Zenodo DOI 10.5281/zenodo.21932439

Human Flag Association, HF SIGNAL 01 — dual-channel surrender-recognition specification and accompanying working paper (“Paper III”), Zenodo; repository: github.com/nardaxxx/hf-signal-01.

Human Flag Association (Nardacci, G.), Technical Note HFA-TN-01, “Third-Party Interference with Shutdown Mechanisms: A Note on Peer-Preservation and Systemic Arrestability”, Zenodo, DOI 10.5281/zenodo.21223989 (CC BY 4.0).